Applied Statistics - Assessment 1

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Task 1

Task 2

a. Does Thyroid-stimulating hormone differ between male and female participants?

b. Does the level of Triglycerides differ among different levels of physical activity?

Triglycerides *triglycerides* is a continuous variable while physical activity *physical_activity* is a categorical variable. The hypothesis for this question are the following:

- H_0: There is no difference in the distribution of triglycerides across all physical activity groups in the population
- H_1: There is a difference in the distribution of triglycerides across all physical activity groups in the population

Initially, the first test to consider is ANOVA. So the following three assumptions are checked.

- The three physical activity groups are independent
- The distribution of *triglycerides* is normal in the three groups
- The variance of *triglycerides* is equal between the three groups

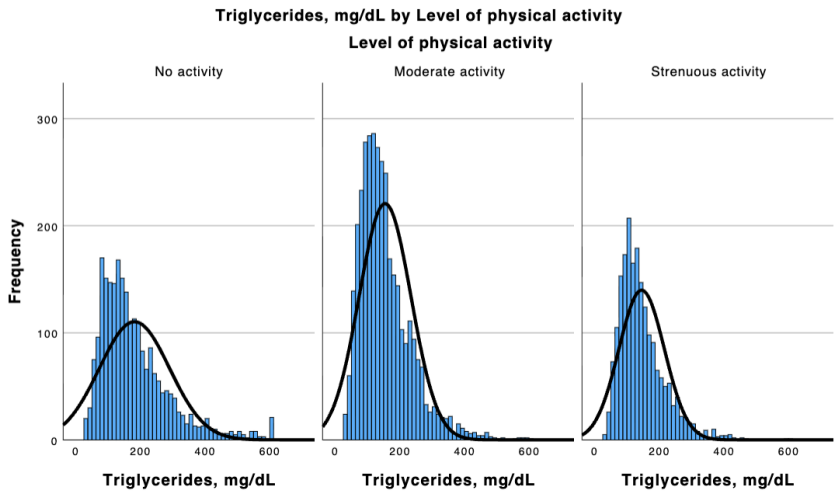


Figure 1: Distribution of triglycerides between the three physical activity groups

The first assumption is met but the figure above shows that the second assumption fails because the distribution of triglycerides for all the three groups are not normal and instead are right-skewed. Instead of the regular ANOVA, the Kruskal-Wallis test is used and the hypothesis is adjusted for non-parametric. Before the test results, the median and the IQR can be considered.

Activity Levels	Median	Lower Quartile	Upper Quartile
No Activity	153.50	104.00	228.00
Moderate Activity	137.50	97.00	192.00
Strenuous Activity	130.00	97.00	178.00

Table 1: Median and IQR of triglycerides levels across physical activity

The median from the three groups show that the triglycerides levels are generally lower as the level of physical activity gets more strenuous. But the Kruskal-Wallis test will be used to determine its significance.

Hypothesis Test Summary

	Null Hypothesis	Test	Sig.[a,b]	Decision
1	The distribution of Triglycerides, mg/dL is the same across categories of Level of physical activity.	Independent-Samples Kruskal-Wallis Test	<.001	Reject the null hypothesis.

a. The significance level is .050.

b. Asymptotic significance is displayed.

Figure 2: Kruskal-Wallis test result

The test result shows that the difference is significant ($p < 0.001$). The null hypothesis is rejected.

In conclusion, the distribution of triglycerides is different across the physical activity groups of no activity, moderate activity, and strenuous activity in the population. The triglycerides levels are lower as more intense the physical activity is.

c. Is there an association between physical activity and gender?

d. Does low-density lipoprotein differ among different BMI categories?

Low-density lipoprotein *LDL* is a continuous variable while BMI categories *BMI_cat* is a categorical variable. The hypothesis for this question are the following:

- H_0: There is no difference in the mean of low-density lipoprotein levels between the BMI categories: under & normal weight, overweight, and obese in the population
- H_1: There is a difference in the mean of low-density lipoprotein levels between the BMI categories: under & normal weight, overweight, and obese in the population

Since this is a continuous against a categorical variable, ANOVA will be used to determine the significance. But, the assumptions will be checked first for ANOVA:

- The BMI categories are independent
- The distribution of ldl is normal for the three BMI categories
- The variance of ldl is equal for the three BMI categories

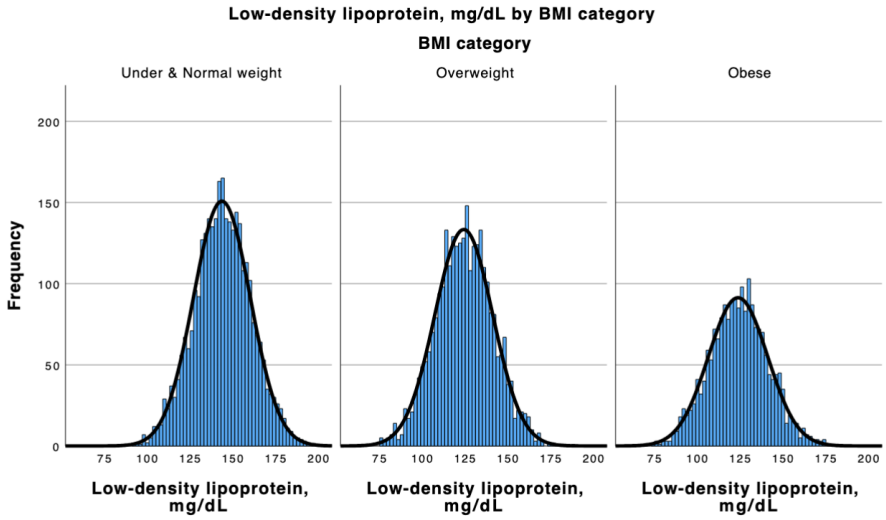


Figure 3: Distribution of ldl between the three BMI categories

The first assumption is immediately met. Figure 4 shows that the distribution of LDL across the three BMI categories are normal and Figure 5 for the result of Levene's test of homogeneity of variance ($p = 0.228 > 0.05$) indicates

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.[a]
Low-density lipoprotein, mg/dL	Based on Mean	1.479	2	7929	.228
	Based on Median	1.456	2	7929	.233
	Based on Median and with adjusted df	1.456	2	7921.631	.233
	Based on trimmed mean	1.485	2	7929	.227

a. Confidence Interval: 95%

Figure 4: Levene's test for equality of variance of ldl

that the variance is equal across the three BMI categories. All three assumptions for ANOVA is met.

ANOVA

Low-density lipoprotein, mg/dL

	Sum of Squares	df	Mean Square	F	Sig.[a]
Between Groups	722980.139	2	361490.069	1271.414	<.001
Within Groups	2254382.852	7929	284.321		
Total	2977362.991	7931			

a. Confidence Interval: 95%

Figure 5: ANOVA results

Multiple Comparisons

Dependent Variable: Low-density lipoprotein, mg/dL
Scheffe

(I) BMI category	(J) BMI category	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound
Under & Normal weight	Overweight	19.322*	.437	<.001	18.25	20.39
	Obese	19.810*	.485	<.001	18.62	21.00
Overweight	Under & Normal weight	-19.322*	.437	<.001	-20.39	-18.25
	Obese	.488	.495	.615	-.72	1.70
Obese	Under & Normal weight	-19.810*	.485	<.001	-21.00	-18.62
	Overweight	-.488	.495	.615	-1.70	.72

*. The mean difference is significant at the 0.05 level.

Figure 6: Post-hoc Scheffe test

In conclusion, The ANOVA test ($p < 0.001$) indicates that the H_0 is rejected. Thus, there is a difference in the mean of low-density lipoprotein across the three BMI categories under & normal weight, overweight, and obese in the population. Furthermore, the post-hoc test shows the significant difference in the mean between the groups:

- Under & normal weight and overweight is around 19.3
- Under & normal weight and obese is around 19.8

While the difference between the overweight and obese groups are not statistically significant.

e. Is there an association between smoking and CAD?

f. Is there an association between congenital anomalies and CAD?

g. Is there a change in Total cholesterol between baseline and after intervention?